№3. : Daisyworld

2026-03-20

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6
7
8 (ramp)
9
10
11 (ramp)
12

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- -01-23
-
- : 1132239110

- :
- : Daisyworld
- : Julia, Agents.jl, DrWatson, CairoMakie, Literate.jl
- : ,

2

-
- Daisyworld `Agents.jl`
-
-

1. Julia- `DrWatson`.
2. Daisyworld.
3. .
4. .
5. `init_white` `max_age`.
6. `script`, `notebook`, `markdown`.

3

(ABM)

-
-
-
- `Agents.jl`

Daisyworld

- :
- ,
-
-

- : $\alpha_b = 0.25$
- : $\alpha_w = 0.75$
- :

$$T = 72\ln((1-\alpha)\cdot L) + 80$$

- :

$$p = 0.1457T - 0.0032T^2 - 0.6443$$

4

Julia

Julia REPL .

```
lilidji@eiglushchenko:~$ cd /home/lilidji/labs/lab03/project
lilidji@eiglushchenko:~/labs/lab03/project$ julia

               _
   _       _ _(_)_     |  Documentation: https://docs.julialang.org
  (_)     | (_) (_)    |
   _ _   _| |_  __ _   |  Type "?" for help, "]?" for Pkg help.
  | | | | | | |/ _` |  |
  | | |_| | | | (_| |  |  Version 1.12.5 (2026-02-09)
 _/ |\__'_|_|_|\__'_|  |  Official https://julialang.org release
|__/                   |
```

Pkg, .

```
julia> using Pkg

julia>

julia> Pkg.activate("/home/lilidji/labs/lab03/project")
  Activating project at `~/labs/lab03/project`

julia> Pkg.add([
           "DrWatson",
           "Agents",
           "CairoMakie",
           "Literate",
           "IJulia",
           "Random",
           "StatsBase",
           "Statistics"
       ])
    Updating registry at `~/.julia/registries/General.toml`
   Resolving package versions...
      Compat entries added for
     Project No packages added to or removed from `~/labs/lab03/project/Project.toml`
    Manifest No packages added to or removed from `~/labs/lab03/project/Manifest.toml`

julia> Pkg.precompile()
```

- DrWatson
- Agents, CairoMakie, Literate, IJulia, StatsBase
-
- src, scripts, plots, notebooks, docs

5

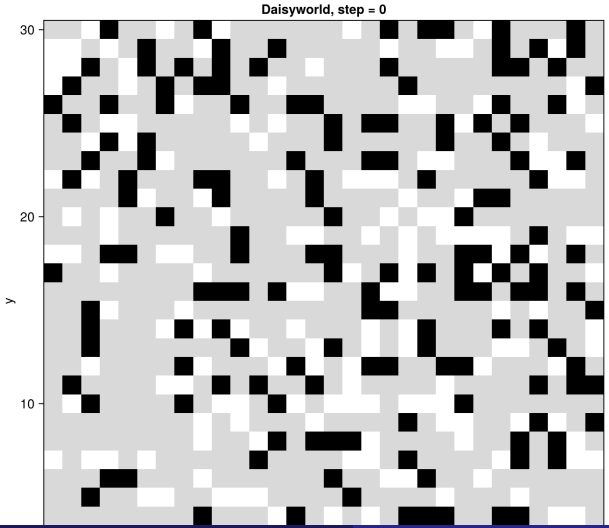
daisyworld.jl

```
lilidji@eiglushchenko:~/labs/lab03/project$ ~/.juliaup/bin/julia --project=. scripts/daisyworld.jl
saved: /home/lilidji/labs/lab03/project/plots/daisy_step000.png
saved: /home/lilidji/labs/lab03/project/plots/daisy_step001.png
saved: /home/lilidji/labs/lab03/project/plots/daisy_step005.png
saved: /home/lilidji/labs/lab03/project/plots/daisy_step040.png
```

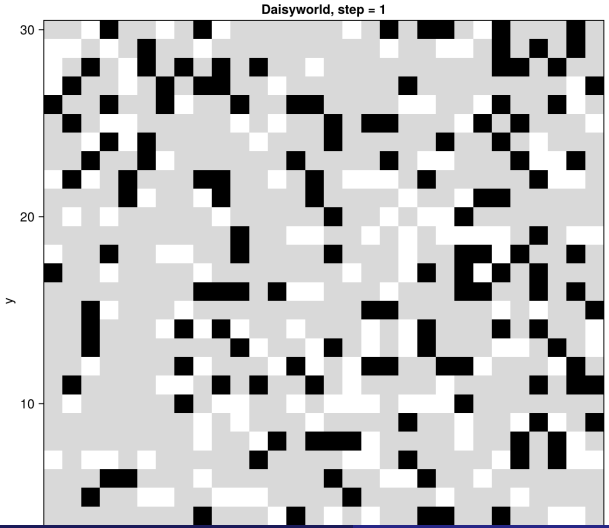
: 0

:

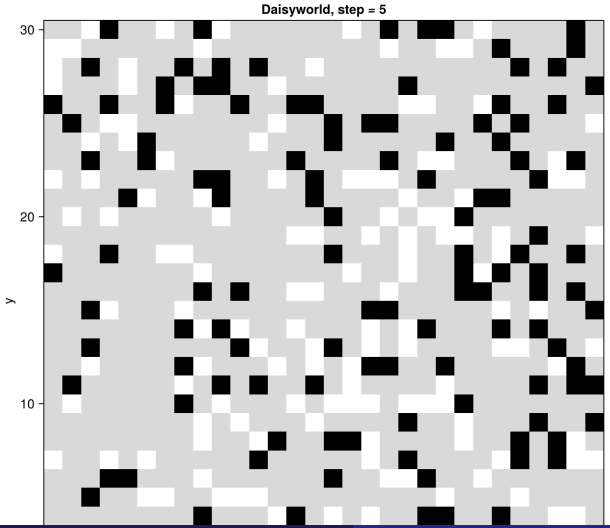
.



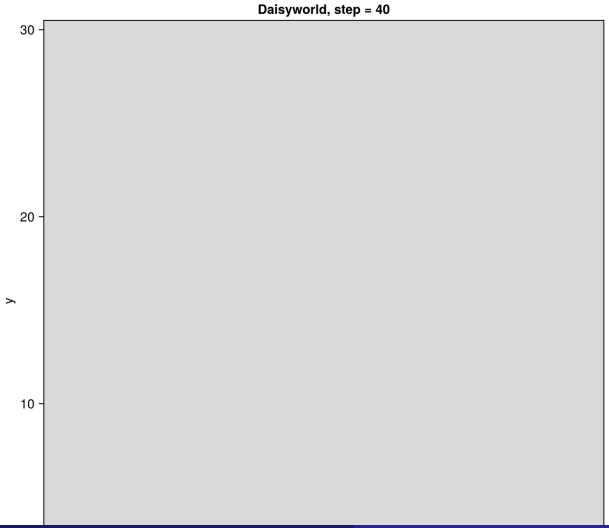
: 1



: 5



: 40



Literate.script: daisyworld

-
- scripts/daisyworld/
-

Literate.notebook: daisyworld

- Jupyter notebook
- notebooks/daisyworld/
-

Literate.markdown: daisyworld

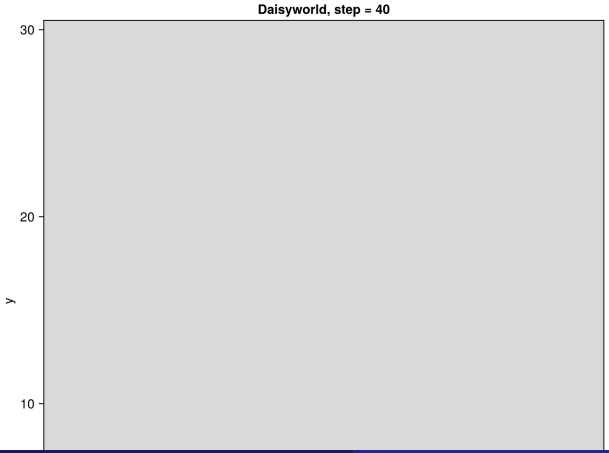
- Markdown-
- docs/daisyworld/
-

6

daisyworld-animate.jl

```
lilidji@eiglushchenko:~/labs/lab03/project$ ~/.juliaup/bin/julia --project=. scripts/daisyworld-animate.jl
saved: /home/lilidji/labs/lab03/project/plots/daisyworld_animation.mp4
```

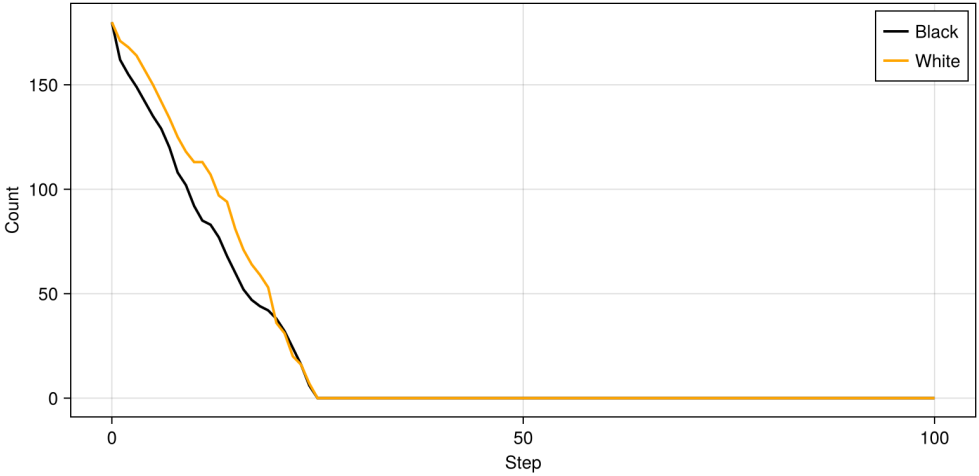
- daisyworld_animation.mp4
-
-



7

daisyworld-count.jl

- 100
-
-


Black
White
150
100
50
0
Count
0
50
100
Step

Literate.script: daisyworld-count

-
- `scripts/daisyworld-count/`
-

Literate.markdown: daisyworld-count

- Markdown-
- docs/daisyworld-count/
-

Literate.notebook: daisyworld-count

- Jupyter notebook
- notebooks/daisyworld-count/
-

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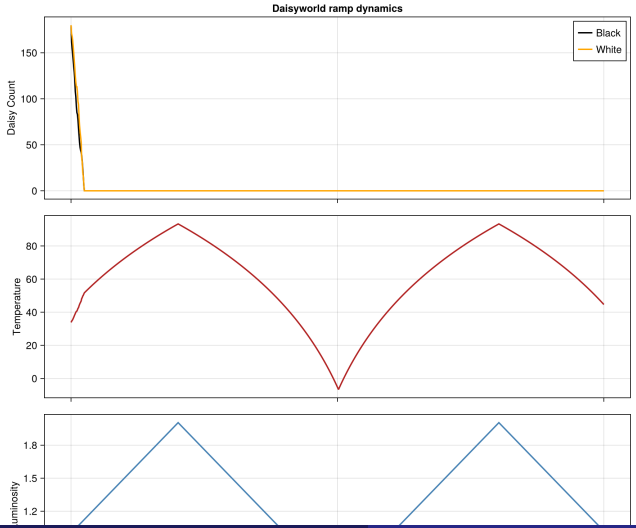
(ramp)

daisyworld-luminosity.jl

- ramp
- , ,
- 1000


Daisyworld ramp dynamics
Black
White
Daisy Count
150
100
50
0
Temperature
80
60
40
20
0
uminosity
1.8
1.5
1.2

Literate.script: daisyworld-luminosity

-
- `scripts/daisyworld-luminosity/`
-

Literate.markdown: daisyworld-luminosity

- Markdown-
- docs/daisyworld-luminosity/
-

Literate.notebook: daisyworld-luminosity

- Jupyter notebook
- notebooks/daisyworld-luminosity/
-

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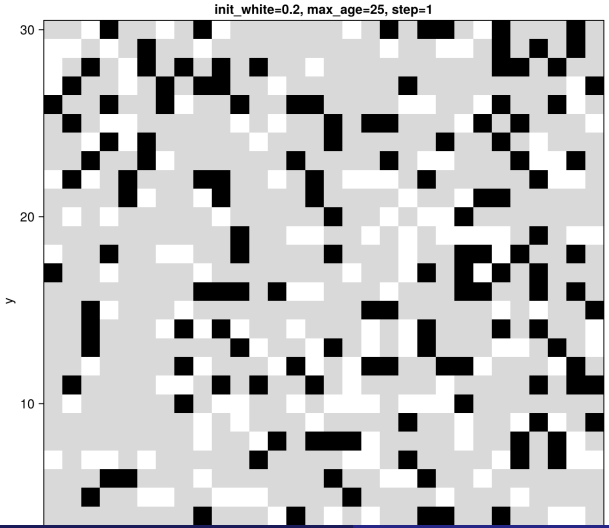
daisyworld_param.jl

:

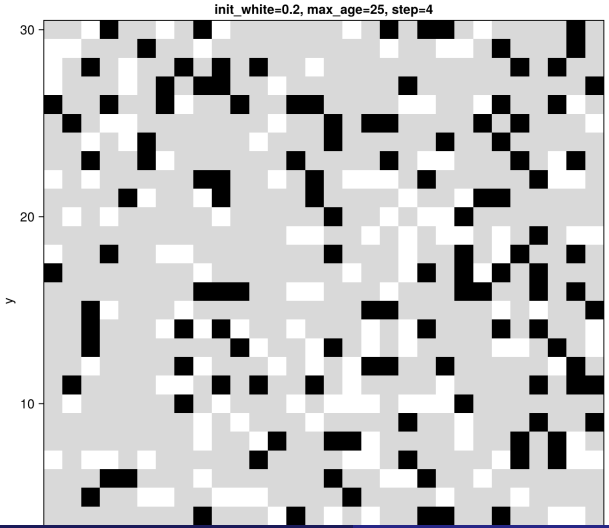
- init_white = 0.2, max_age = 25
- init_white = 0.2, max_age = 40
- init_white = 0.8, max_age = 25
- init_white = 0.8, max_age = 40

```
lilidji@eiglushchenko:~/labs/lab03/project$ ~/.juliaup/bin/julia --project=. scripts/daisyworld_param.jl
saved: /home/lilidji/labs/lab03/project/plots/daisyworld_init_white=0.2_max_age=25.png_step00.png
saved: /home/lilidji/labs/lab03/project/plots/daisyworld_init_white=0.2_max_age=25.png_step01.png
saved: /home/lilidji/labs/lab03/project/plots/daisyworld_init_white=0.2_max_age=25.png_step04.png
saved: /home/lilidji/labs/lab03/project/plots/daisyworld_init_white=0.2_max_age=25.png_step40.png
saved: /home/lilidji/labs/lab03/project/plots/daisyworld_init_white=0.2_max_age=40.png_step00.png
saved: /home/lilidji/labs/lab03/project/plots/daisyworld_init_white=0.2_max_age=40.png_step01.png
saved: /home/lilidji/labs/lab03/project/plots/daisyworld_init_white=0.2_max_age=40.png_step04.png
saved: /home/lilidji/labs/lab03/project/plots/daisyworld_init_white=0.2_max_age=40.png_step40.png
saved: /home/lilidji/labs/lab03/project/plots/daisyworld_init_white=0.8_max_age=25.png_step00.png
saved: /home/lilidji/labs/lab03/project/plots/daisyworld_init_white=0.8_max_age=25.png_step01.png
saved: /home/lilidji/labs/lab03/project/plots/daisyworld_init_white=0.8_max_age=25.png_step04.png
saved: /home/lilidji/labs/lab03/project/plots/daisyworld_init_white=0.8_max_age=25.png_step40.png
saved: /home/lilidji/labs/lab03/project/plots/daisyworld_init_white=0.8_max_age=40.png_step00.png
saved: /home/lilidji/labs/lab03/project/plots/daisyworld_init_white=0.8_max_age=40.png_step01.png
saved: /home/lilidji/labs/lab03/project/plots/daisyworld_init_white=0.8_max_age=40.png_step04.png
saved: /home/lilidji/labs/lab03/project/plots/daisyworld_init_white=0.8_max_age=40.png_step40.png
```

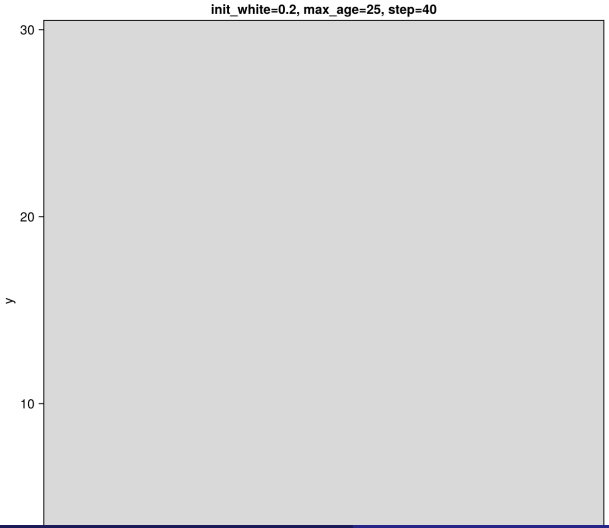
init_white = 0.2, max_age = 25: 1



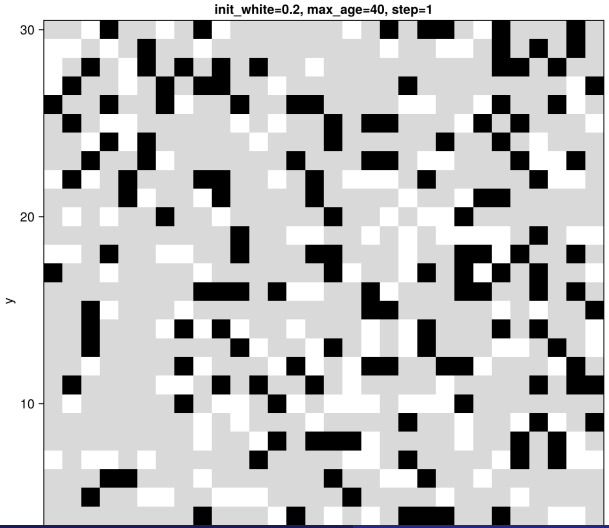
init_white = 0.2, max_age = 25: 4



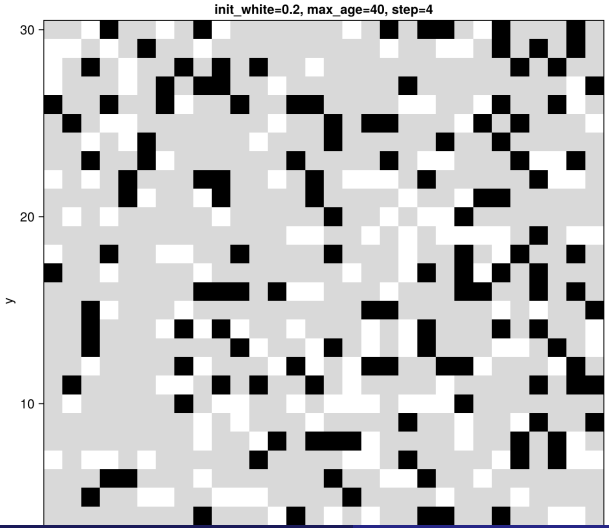
init_white = 0.2, max_age = 25: 40



init_white = 0.2, max_age = 40: 1

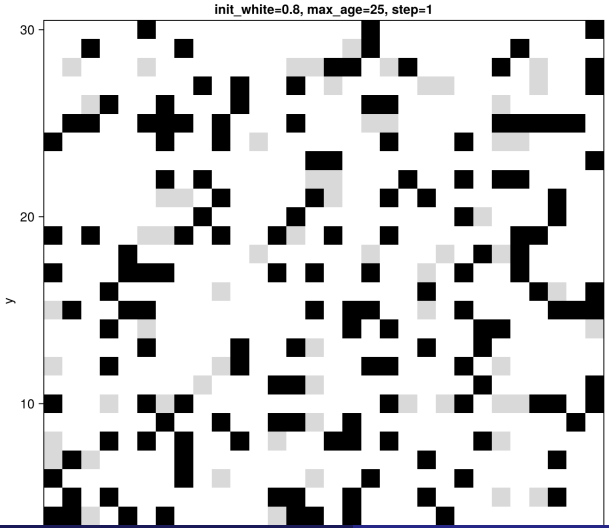


init_white = 0.2, max_age = 40: 4

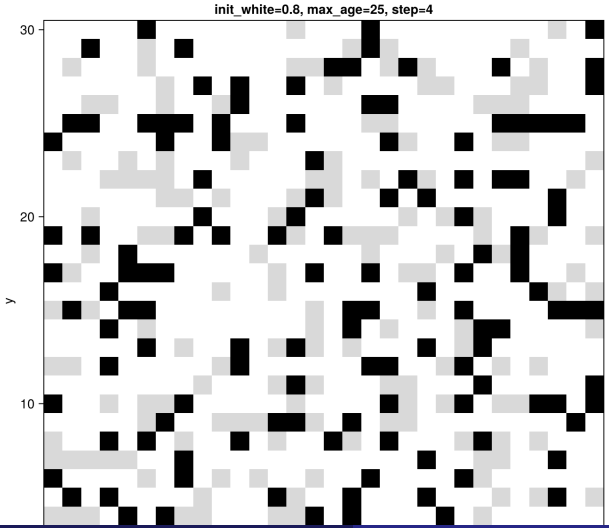


`init_white = 0.2, max_age = 40:` 40

init_white = 0.8, max_age = 25: 1



init_white = 0.8, max_age = 25: 4



init_white = 0.8, max_age = 25: 40

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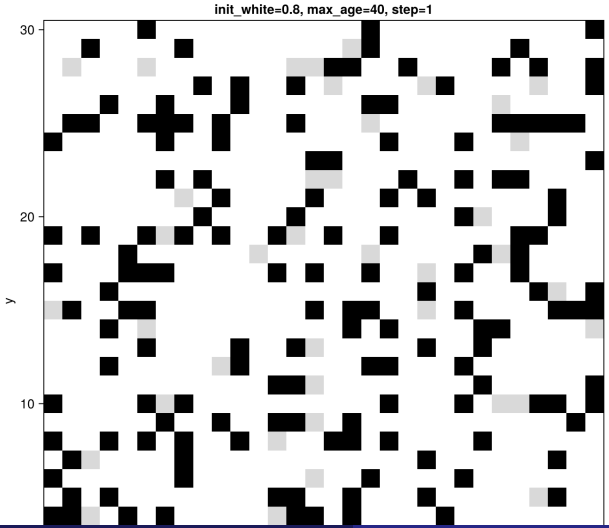
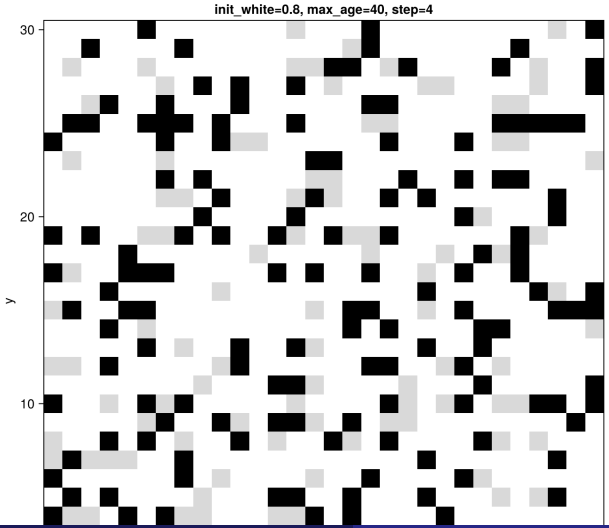
init_white = 0.8, max_age = 40: 1



init_white = 0.8, max_age = 40: 4



init_white = 0.8, max_age = 40: 40

Literate.script: daisyworld_param

- heatmap
- scripts/daisyworld_param/
-

Literate.markdown: daisyworld_param

- Markdown-
- docs/daisyworld_param/
-

Literate.notebook: daisyworld_param

- Jupyter notebook
- notebooks/daisyworld_param/
-

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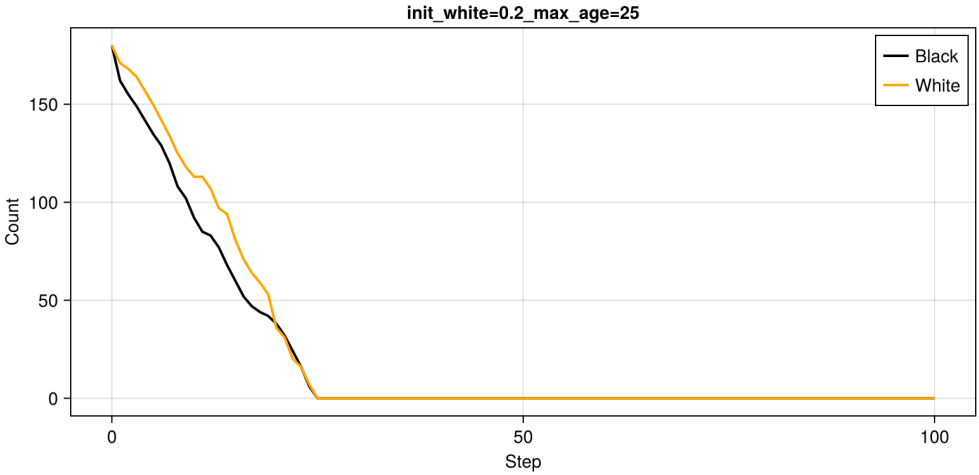
daisyworld-count__param.jl

```
saved: /home/lilidji/labs/lab03/project/plots/daisyworld_init_white=0.8_max_age=40.png_step40.png
lilidji@eiglushchenko:~/labs/lab03/project$ ~/.juliaup/bin/julia --project=. scripts/daisyworld-count__param.jl
saved: /home/lilidji/labs/lab03/project/plots/daisy-count_init_white=0.2_max_age=25.png
saved: /home/lilidji/labs/lab03/project/plots/daisy-count_init_white=0.2_max_age=40.png
saved: /home/lilidji/labs/lab03/project/plots/daisy-count_init_white=0.8_max_age=25.png
saved: /home/lilidji/labs/lab03/project/plots/daisy-count_init_white=0.8_max_age=40.png
lilidji@eiglushchenko:~/labs/lab03/project$
```

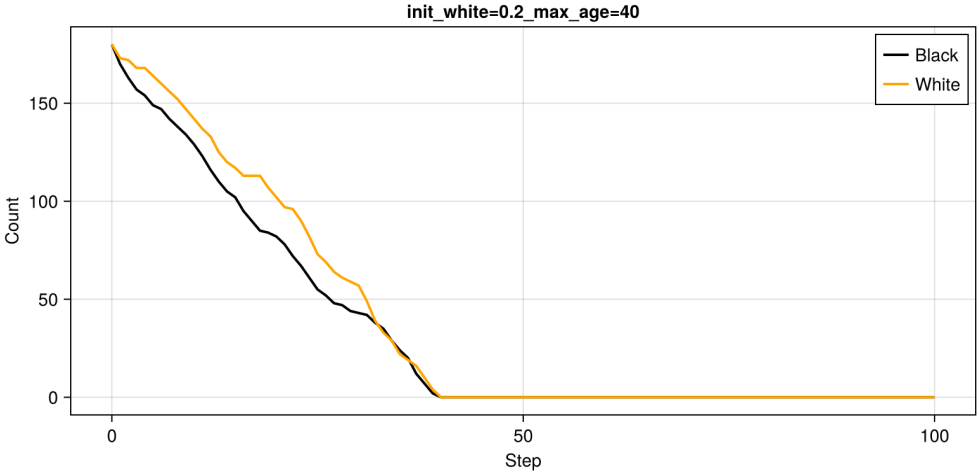
init_white = 0.2, max_age = 25:

.



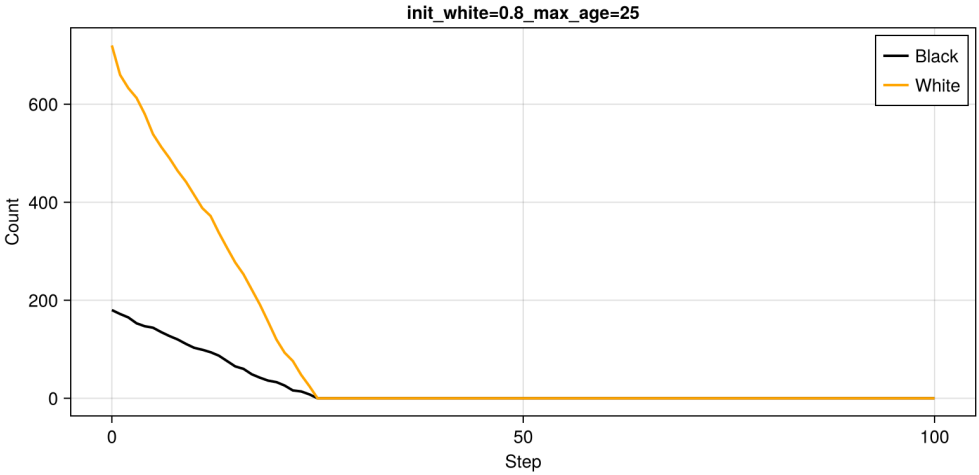
init_white = 0.2, max_age = 40:

max_age .



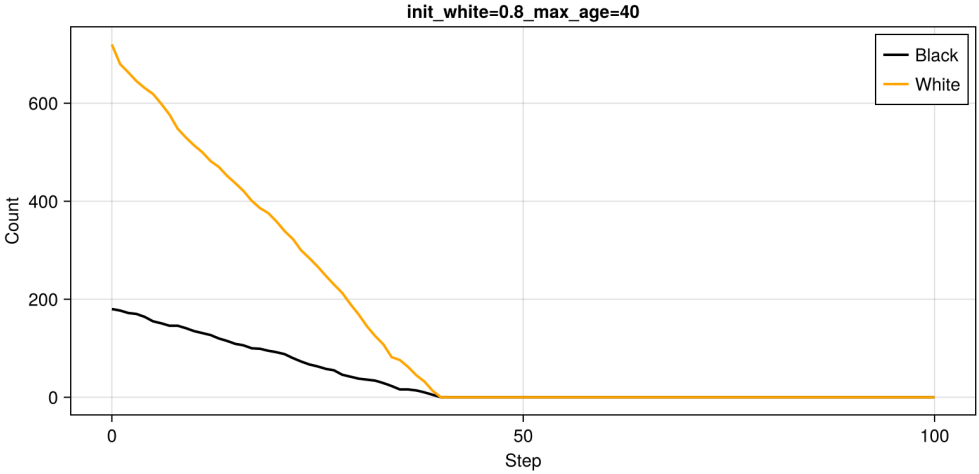
init_white = 0.8, max_age = 25:

.



init_white = 0.8, max_age = 40:

.



Literate.script: daisyworld-count__param

-
- scripts/daisyworld-count__param/
-

Literate.markdown: daisyworld-count__param

- Markdown-
- docs/daisyworld-count__param/
-

Literate.notebook: daisyworld-count__param

- Jupyter notebook
- notebooks/daisyworld-count__param/
-

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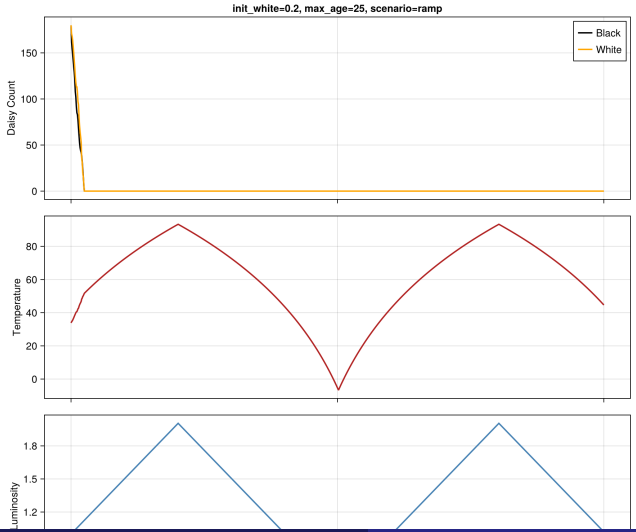
(ramp)

daisyworld-luminosity__param.jl

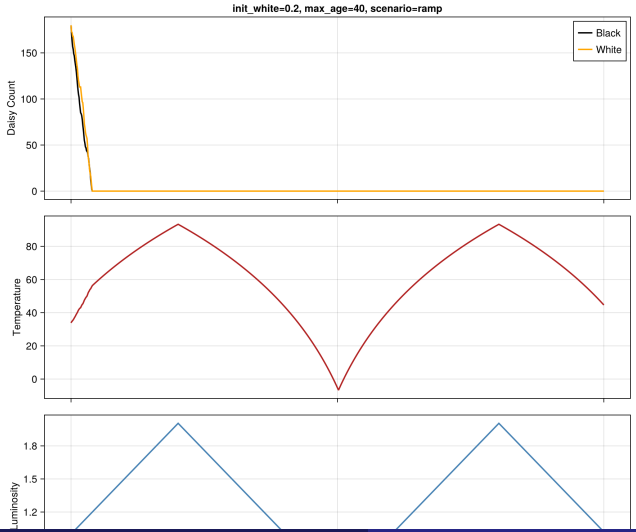
ramp.

```
saved: /home/lilidji/labs/lab03/project/plots/daisy-count_init_white=0.8_max_age=40.png
lilidji@eiglushchenko:~/labs/lab03/project$ ~/.juliaup/bin/julia --project=. scripts/daisyworld-luminosity__par
saved: /home/lilidji/labs/lab03/project/plots/daisy-luminosity_init_white=0.2_max_age=25_scenario=ramp.png
saved: /home/lilidji/labs/lab03/project/plots/daisy-luminosity_init_white=0.2_max_age=40_scenario=ramp.png
saved: /home/lilidji/labs/lab03/project/plots/daisy-luminosity_init_white=0.8_max_age=25_scenario=ramp.png
saved: /home/lilidji/labs/lab03/project/plots/daisy-luminosity_init_white=0.8_max_age=40_scenario=ramp.png
lilidji@eiglushchenko:~/labs/lab03/project$
```

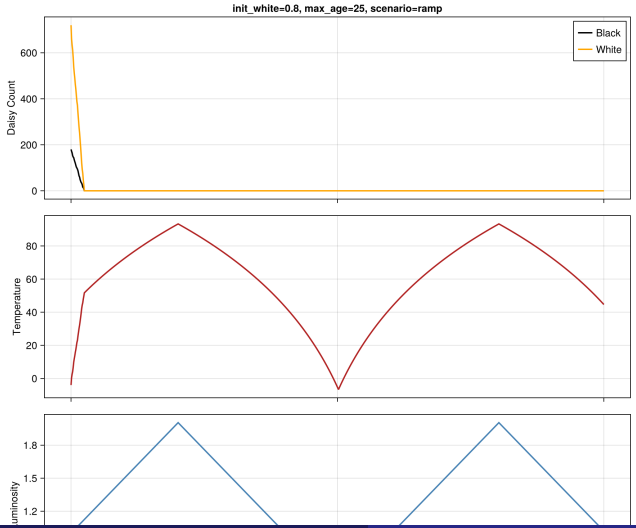
init_white = 0.2, max_age = 25 (ramp)



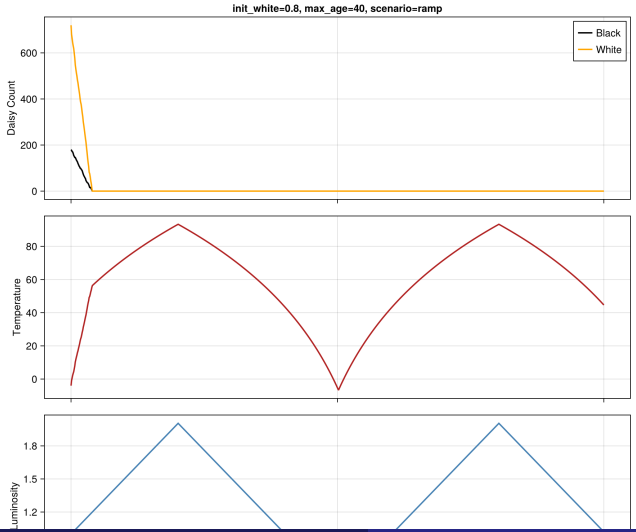
init_white = 0.2, max_age = 40 (ramp)



init_white = 0.8, max_age = 25 (ramp)



init_white = 0.8, max_age = 40 (ramp)



Literate.script: daisyworld-luminosity__param

-
- scripts/daisyworld-luminosity__param/
-

Literate.markdown: daisyworld-luminosity__param

- Markdown-
- docs/daisyworld-luminosity__param/
- ramp

Literate.notebook: daisyworld-luminosity__param

- Jupyter notebook
- notebooks/daisyworld-luminosity__param/
-

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tangle.jl,

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```
lilidji@eiglushchenko:~/labs/lab03/project$ ~/.juliaup/bin/julia --project=. scripts/tangle.jl scripts/daisyworld.jl
[ Info: generating plain script file from `~/labs/lab03/project/scripts/daisyworld.jl`
[ Info: writing result to `~/labs/lab03/project/scripts/daisyworld/daisyworld.jl`
[ Info: generating notebook from `~/labs/lab03/project/scripts/daisyworld.jl`
[ Info: writing result to `~/labs/lab03/project/notebooks/daisyworld/daisyworld.ipynb`
[ Info: generating markdown page from `~/labs/lab03/project/scripts/daisyworld.jl`
[ Info: writing result to `~/labs/lab03/project/docs/daisyworld/daisyworld.md`
generated for daisyworld
  script   -> /home/lilidji/labs/lab03/project/scripts/daisyworld/daisyworld.jl
  notebook -> /home/lilidji/labs/lab03/project/notebooks/daisyworld/daisyworld.ipynb
  markdown -> /home/lilidji/labs/lab03/project/docs/daisyworld/daisyworld.md
lilidji@eiglushchenko:~/labs/lab03/project$ ~/.juliaup/bin/julia --project=. scripts/tangle.jl scripts/daisyworld-count.jl
 ~/.juliaup/bin/julia --project=. scripts/tangle.jl scripts/daisyworld-luminosity.jl
 ~/.juliaup/bin/julia --project=. scripts/tangle.jl scripts/daisyworld__param.jl
 ~/.juliaup/bin/julia --project=. scripts/tangle.jl scripts/daisyworld-count__param.jl
 ~/.juliaup/bin/julia --project=. scripts/tangle.jl scripts/daisyworld-luminosity__param.jl
[ Info: generating plain script file from `~/labs/lab03/project/scripts/daisyworld-count.jl`
[ Info: writing result to `~/labs/lab03/project/scripts/daisyworld-count/daisyworld-count.jl`
[ Info: generating notebook from `~/labs/lab03/project/scripts/daisyworld-count.jl`
[ Info: writing result to `~/labs/lab03/project/notebooks/daisyworld-count/daisyworld-count.ipynb`
[ Info: generating markdown page from `~/labs/lab03/project/scripts/daisyworld-count.jl`
[ Info: writing result to `~/labs/lab03/project/docs/daisyworld-count/daisyworld-count.md`
generated for daisyworld-count
  script   -> /home/lilidji/labs/lab03/project/scripts/daisyworld-count/daisyworld-count.jl
  notebook -> /home/lilidji/labs/lab03/project/notebooks/daisyworld-count/daisyworld-count.ipynb
  markdown -> /home/lilidji/labs/lab03/project/docs/daisyworld-count/daisyworld-count.md
[ Info: generating plain script file from `~/labs/lab03/project/scripts/daisyworld-luminosity.jl`
[ Info: writing result to `~/labs/lab03/project/scripts/daisyworld-luminosity/daisyworld-luminosity.jl`
[ Info: generating notebook from `~/labs/lab03/project/scripts/daisyworld-luminosity.jl`
[ Info: writing result to `~/labs/lab03/project/notebooks/daisyworld-luminosity/daisyworld-luminosity.ipynb`
[ Info: generating markdown page from `~/labs/lab03/project/scripts/daisyworld-luminosity.jl`
[ Info: writing result to `~/labs/lab03/project/docs/daisyworld-luminosity/daisyworld-luminosity.md`
generated for daisyworld-luminosity
  script   -> /home/lilidji/labs/lab03/project/scripts/daisyworld-luminosity/daisyworld-luminosity.jl
  notebook -> /home/lilidji/labs/lab03/project/notebooks/daisyworld-luminosity/daisyworld-luminosity.ipynb
  markdown -> /home/lilidji/labs/lab03/project/docs/daisyworld-luminosity/daisyworld-luminosity.md
[ Info: generating plain script file from `~/labs/lab03/project/scripts/daisyworld__param.jl`
[ Info: writing result to `~/labs/lab03/project/scripts/daisyworld__param/daisyworld__param.jl`
[ Info: generating notebook from `~/labs/lab03/project/scripts/daisyworld__param.jl`
[ Info: writing result to `~/labs/lab03/project/notebooks/daisyworld__param/daisyworld__param.ipynb`
[ Info: generating markdown page from `~/labs/lab03/project/scripts/daisyworld__param.jl`
[ Info: writing result to `~/labs/lab03/project/docs/daisyworld__param/daisyworld__param.md`
generated for daisyworld__param
  script   -> /home/lilidji/labs/lab03/project/scripts/daisyworld__param/daisyworld__param.jl
  notebook -> /home/lilidji/labs/lab03/project/notebooks/daisyworld__param/daisyworld__param.ipynb
  markdown -> /home/lilidji/labs/lab03/project/docs/daisyworld__param/daisyworld__param.md
[ Info: generating plain script file from `~/labs/lab03/project/scripts/daisyworld-count__param.jl`
[ Info: writing result to `~/labs/lab03/project/scripts/daisyworld-count__param/daisyworld-count__param.jl`
[ Info: generating notebook from `~/labs/lab03/project/scripts/daisyworld-count__param.jl`
[ Info: writing result to `~/labs/lab03/project/notebooks/daisyworld-count__param/daisyworld-count__param.ipynb`
[ Info: generating markdown page from `~/labs/lab03/project/scripts/daisyworld-count__param.jl`
[ Info: writing result to `~/labs/lab03/project/docs/daisyworld-count__param/daisyworld-count__param.md`
generated for daisyworld-count__param
  script   -> /home/lilidji/labs/lab03/project/scripts/daisyworld-count__param/daisyworld-count__param.jl
  notebook -> /home/lilidji/labs/lab03/project/notebooks/daisyworld-count__param/daisyworld-count__param.ipynb
```

:

- plots
- scripts/<name>/
- notebooks/<name>/
- Markdown- docs/<name>/

```
 markdown -> /home/lilldjl/labs/lab03/project/docs/daisyworld-luminosity__param/daisyworld-luminosity__param.m
lilldjl@eiglushchenko:~/labs/lab03/project$ find plots -maxdepth 1 -type f | sort
find scripts -maxdepth 2 -type f | sort
find notebooks -maxdepth 2 -type f | sort
find docs -maxdepth 2 -type f | sort
plots/daisy-count_init_white=0.2_max_age=25.png
plots/daisy-count_init_white=0.2_max_age=40.png
plots/daisy-count_init_white=0.8_max_age=25.png
plots/daisy-count_init_white=0.8_max_age=40.png
plots/daisy-luminosity_init_white=0.2_max_age=25_scenario=ramp.png
plots/daisy-luminoslty_init_white=0.2_max_age=40_scenario=ramp.png
plots/daisy-luminosity_init_white=0.8_max_age=25_scenario=ramp.png
plots/daisy-luminosity_init_white=0.8_max_age=40_scenario=ramp.png
plots/daisy_step000.png
plots/daisy_step001.png
plots/daisy_step005.png
plots/daisy_step040.png
plots/daisyworld_animation.mp4
plots/daisyworld_init_white=0.2_max_age=25.png_step00.png
plots/daisyworld_init_white=0.2_max_age=25.png_step01.png
plots/daisyworld_init_white=0.2_max_age=25.png_step04.png
plots/daisyworld_init_white=0.2_max_age=25.png_step40.png
plots/daisyworld_init_white=0.2_max_age=40.png_step00.png
plots/daisyworld_init_white=0.2_max_age=40.png_step01.png
plots/daisyworld_init_white=0.2_max_age=40.png_step04.png
plots/daisyworld_init_white=0.2_max_age=40.png_step40.png
plots/daisyworld_init_white=0.8_max_age=25.png_step00.png
plots/daisyworld_init_white=0.8_max_age=25.png_step01.png
plots/daisyworld_init_white=0.8_max_age=25.png_step04.png
plots/daisyworld_init_white=0.8_max_age=25.png_step40.png
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plots/daisyworld_init_white=0.8_max_age=40.png_step01.png
plots/daisyworld_init_white=0.8_max_age=40.png_step04.png
```

- Daisyworld Julia `Agents.jl`
- ,
- `init_white`
- `max_age`
- `ramp`
- `Literate.jl`